Remote Controller Operator Reference

All editable parameters & the full control interface

*Team **To Be Defined** — SDC26*

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Contents

1 Introduction & architecture

The Remote Web Controller (**`controller/remote_web_controller.py`**, default port **8090**) is the single operator cockpit for the Team To Be Defined drone swarm. It proxies commands to per-drone flight controllers (**`controller_unified/unified_api_server.py`**, port **8080** on each Raspberry Pi) and exposes every tunable knob, every pose-tracking parameter, and every mission workflow in one browser page.

The UI is a single self-contained HTML/JS page built from the Python file. All state is queried via HTTP (drone config, telemetry, position) and pushed via HTTP POST or a WebSocket-free SSE channel for positions. A single operator can switch between up to 5 Anafi drones by number-key or by clicking the drone bar.

2 Starting the controller

On the control laptop (the C2):

```
cd sdc-tobe/controller
python3 remote_web_controller.py
```

Open **`http://localhost:8090`** in a modern browser. The page starts empty until it polls **`/proxy/drones`** and paints the drone bar. Each Raspberry Pi must already be running its per-drone **`unified_api_server.py`** on port 8080.

> ***Tip —*** *If the drone bar stays empty, verify the Pi's* **`flightctrlN`** *hostname resolves (from* **`controller/drones_config.json`***) and that the Pi's API answers* **`GET /ping`***.*

3 UI layout at a glance

From top to bottom the page is organised into collapsible sections. Each section remembers its collapsed state in *localStorage* so each operator gets the layout they left the last session with.

UI control	What it does
Header row	Drone selector, LAND ALL, Config, theme toggle, logo.
Latency widget	C2→FC + FC→drone RTT plus video offset; auto-applies total to Position Tracker latency.
Tuning Parameters	The single home for every live-tunable knob — observer PD and position tracker side-by-side.
WASD Key controls	The 3×3 key grid + takeoff/land/recover.
Anafi / Olympe controls	Gimbal, magnetometer, environment, Wi-Fi, camera zoom, altitude/speed/tilt limits.
Video stream	MJPEG or UDP-forward live feed, with zoom slider.
Telemetry	Battery, GPS, altitude, attitude, velocity, state.
ArUco Seek	Visual-servo hover panel (observer/live mode).
Mission Planner	Free-form step list (takeoff, move, wait, land).
Special Missions	Pre-built missions: Scan-all, Capture-targets.
Position Tracker	Arena-frame 2D+3D views with live pose fusion.
Arena Configuration	Marker layout editor & physical dims.

4 Header row

UI control	What it does
Drone bar	Row of drone buttons (one per entry in `drones_config.json`). Click to switch active drone; hotkeys 1–5 do the same.
PAUSE ALL (9)	Amber override. Freezes every drone mid-air at its current position — missions abort, ArUco Seek drops to OBSERVE, a zero-RC brake goes out. Only manual WASD is active after this. See §5a for full behaviour.
CONTINUE MISSION	Appears only while paused. Clears the flag so autonomous endpoints become callable again. Nothing auto-restarts — the operator must re-arm any mission manually.
LAND ALL (0)	Red emergency button. Lands every drone in the fleet. Keyboard shortcut is the zero key (not in the movement map, safe to hit one-handed).
Config	Opens the Drone Fleet configuration modal. Add / remove / rename drones, change their base URL. Saves to `drones_config.json`.
Theme toggle	■■ Light ↔ ■ Dark. Persisted in localStorage.key `sdc_theme`.
Team logo	Served from `/logo.png` (alpha-masked variant if present, falls back to the original).

4a PAUSE ALL — behaviour detail

PAUSE ALL is a soft kill-switch that leaves the drones flying but locks out autonomous control. Use it the moment something looks wrong but you don't want the violence of a forced landing. The order of operations when you press the button (or hit **9**) is:

UI control	What it does
1. Raise global flag	`_global_paused = True`. Mission-start and ArUco-LIVE endpoints start returning 409 *paused* so nothing can re-arm behind your back.
2. Abort mission	Any running mission is stopped with **land=False** — the FSM tears down and the drone is left hovering.
3. Observers → OBSERVE	Every ArUco Seek observer switches out of LIVE and its search-RC overrides are cleared. The visual-servo loop stops sending RC.
4. Zero-RC brake	A `POST /api/rc {lr:0,fb:0,ud:0,yaw:0}` goes to every drone in parallel. Anafi firmware holds position with no input, so this is an effective air-brake.
5. UI banner	Amber pulsing banner appears; **PAUSE ALL** hides and **CONTINUE MISSION** appears in its place. The Mission Planner, Special Missions, and ArUco Seek panels dim + become non-interactive (CSS `body.paused-mode`).
What still works	WASD / Q-E / R-F / X / Space — i.e. keyboard manual control. The /proxy/rc, /proxy/key_down, /proxy/key_up routes are NOT guarded. Takeoff / Land / LAND ALL also still work — they're emergency-level controls.
What is blocked	/proxy/missions/scan_all/start · /proxy/missions/capture_targets/start · /proxy/aruco/mode when requested=live. Anything else passes through unchanged.
Tab sync	Every open UI tab polls `/proxy/pause_status` every 2 s, so pausing from one tab is reflected in all others.
Resume	CONTINUE MISSION (or hit **9** again) POSTs `/proxy/resume_all`. Flag clears immediately but **no mission auto-restarts** — the operator must re-arm via the Mission Planner / Special Missions panel.

5 Latency widget

Sits below the drone bar. Shows four numbers and one checkbox:

UI control	What it does
Total	Sum of the three components. Green when < 100 ms, amber 100–250, red > 250.
c2 → fc	Round-trip time from the control laptop to the Pi's unified API, measured live by pinging **`/ping`**.
fc → drone	Round-trip time from the Pi to the Anafi's Wi-Fi IP (**`192.168.42.1`**), measured once per second via ICMP ping and cached on the Pi.
video + N ms	User-set constant accounting for local frame processing/decoding on the C2. Not measured — adjust to taste.
auto-set latency	When checked, the total is pushed into the Position Tracker **Latency ms** slider on every poll so IMU rewinding always matches the current link conditions.

6 Keyboard shortcuts

The keydown/keyup handler is active whenever no text field is focused. Keys repeat at ~20 Hz while held.

Key(s)	Action
`W / S`	Pitch forward / back (body-frame X).
`A / D`	Strafe left / right (body-frame Y).
`Q / E`	Rotate left / right (yaw).
`R / F`	Climb / descend (vertical thrust).
`X or Space`	Hard stop — zero all RC axes.
`T`	Takeoff on the active drone.
`L`	Land the active drone.
`9`	**PAUSE / CONTINUE** — freezes the whole fleet in place (autonomous missions abort; only WASD remains active). Press again to resume.
`0`	**Panic: LAND ALL** — lands every drone in the fleet.
`1 – 5`	Switch active drone to fleet slot N.

***Safety** — the keydown handler does not re-fire while a key is held; the Pi sends a 200 ms RC keep-alive on its own. If the browser tab loses focus, all keys are treated as released.*

7 WASD & flight grid

A 3×3 button grid that mirrors the keyboard map — useful for touchscreens. The centre button is **STOP**. Each button press-and-holds the corresponding RC axis while the mouse is down.

8 Takeoff, land, recovery, logging

UI control	What it does
Takeoff (T)	Command Anafi takeoff. Pre-takeoff diagnostics print on server stdout if state is unsafe.
Land (L)	Command soft landing with obstacle avoidance.
Recover	Attempts flat-trim + motor recovery after a crash.
Safe Takeoff: OFF	Toggle. When ON, takeoff only proceeds if all sensors (magneto, GPS, altimeter, alert state) report healthy.

Enable Telemetry Log	Start/stop recording telemetry to `remote_telemetry_log.jsonl`.
Download Telemetry Log	Serve the current JSONL file.
Clear Telemetry Log	Truncate the file.
Command Logging: OFF	Start/stop recording every outgoing command with timestamps. High-frequency events (key_down/key_up) throttled to 2 Hz.
Download Command Log	Serve `remote_command_log.jsonl`.
Clear Command Log	Truncate it.

9 Advanced SDK controls

Collapsed by default. Exposes raw Olympe/SDK operations for manual poking while debugging:

UI control	What it does
Rotate CW 45° / CCW 45°	Fixed yaw step.
Up 30cm / Down 30cm	Fixed vertical translate.
Forward / Back / Left / Right 30cm	Fixed horizontal translate (body-frame).
Stream ON / Stream OFF	Raw stream command (debug only — normal video is started from the Video panel).
Set Speed	Applies the speed-input value (10–100) as the max horizontal speed cap.
Raw SDK input + Send	Send any SDK command string verbatim (e.g. `battery?`, `speed?`).

10 Anafi / Olympe panel

Only shown for drones whose type is *anafi*. Groups Anafi-specific controls that have no Tello equivalent:

UI control	What it does
Gimbal tilt slider	-90° (straight down) to +30° (up). Apply with Set, or jump to Down / Forward.
Magnetometer	Status line + **Recalibrate Magnetometer** button opens the figure-8 wizard (rotate the drone through three axes).
Max altitude (m)	Firmware-enforced ceiling. 0.5–150, default 5.
Max vert spd (m/s)	Firmware-enforced vertical speed cap. 0.1–4, default 0.5.
Max tilt (°)	Firmware-enforced attitude tilt cap. 1–35, default 15.
Apply Settings	POST the three caps together (applied in one command).
Environment	Indoor (GPS-less, relaxed checks) vs Outdoor (GPS-required). Required before takeoff in the hall.
Wi-Fi band / channel	Scan + switch between 2.4 GHz and 5 GHz. Use 5 GHz in busy venues — the Anafi is 5-GHz capable.
Camera zoom	0% – 100% slider mapped onto the Anafi's digital+optical zoom. Live update; no arming.

11 Video stream

UI control	What it does
Mode: Off	No video.

Mode: MJPEG (Way 1)	The Pi decodes the drone's H.264 stream and re-encodes as MJPEG for the browser. Single hop, most compatible.
Mode: UDP forward (Way 2)	Pi forwards raw UDP to C2 for browser-side decode. Lower CPU on Pi, more latency spread.
Start Video / Stop Video	Toggle button.
Video URL	Shows the current MJPEG or forward URL — useful for VLC debugging.

12 Telemetry panel

Scrolling monospace block polled every ~250 ms:

battery %, board temperature, TOF/barometer altitude, flight-time, horizontal/vertical speed, Wi-Fi SNR, attitude pitch/roll/yaw, velocity vx/vy/vz (cm/s), acceleration ax/ay/az (cm/s²), SDK version, serial, GPS lat/lon/alt, gimbal pry, connected flag.

***Battery meter —** green ≥ 70%, amber 35–69%, red < 35%. A drone flashes amber in the drone bar when < 20%.*

13 Position Tracker

Arena-frame pose fusion — uses the drone's camera + ArUco markers + onboard IMU to estimate **x, y, z, heading** in arena coordinates. Feeds every other system: ArUco Seek, mission boundary guard, arena view, collision avoidance.

UI control	What it does
Enable checkbox	Starts/stops the positioner on the Pi and opens/closes the SSE event stream.
3D view (Three.js)	**ON by default.** Shows orbit-able arena with the drone as a small axes gizmo. Uncheck to hide. 2D top-down canvas stays visible alongside the 3D view.
Show all drones	When checked, every fleet drone is plotted; unchecked shows only the active drone.
Coordinate readout	X (cyan) · Y (green) · Z (orange) · heading · velocity · ref count · FPS.
Vx/Vy/Vz, Ax/Ay/Az	Body-frame velocity and acceleration from the IMU telemetry.
Profile / FOV / Latency ms / ArUco ↔ IMU blend	Top config row — most common knobs. See parameter reference §19.
Filters row	Kalman on/off, Marker size (m), Top-K, Outlier (m) — apply with **Apply filters**.
Precision row (advanced)	pose-hold, min refs / min-ref-weight, meas_blend min/max, vel_blend, max Δt, Kalman Q/R. Apply with **Apply precision**.
Apply Config	Pushes the top row (Profile / FOV / Latency) to the server.
Upload Calibration (.npz)	Load a pre-computed intrinsics file from OpenCV calibration. Overrides FOV.
Show ArUco Video	Annotated MJPEG feed with detected markers, rejection reasons, and fused-pose overlay.
Record	Writes an .mp4 of the annotated feed to `controller_unified/recordings/`.
Raw	Record the raw frame instead of the annotated overlay.

14 ArUco Seek

Visual-servo hover panel. The observer continuously runs a PD loop on whichever marker the mission has set as target (or whichever marker has the highest confidence in passive mode). Two modes:

UI control	What it does
OBSERVE	Render + tune — no RC output sent to the drone.
LIVE	Panel writes RC commands to the drone. Requires arming (red banner pulses while armed).
Takeoff / Land	Per-drone shortcuts right in the panel.
STOP RC	Zero all RC axes immediately. Does not land.
Emergency	Cut motors. Drone will fall — use only to avoid imminent destruction.
Reload	Re-fetch observer params from the Pi (useful after manual JSON edits).

15 Mission Planner & Special Missions

Two mission runners sit side-by-side. **Mission Planner** accepts a free-form step list (takeoff, 100 forward, 90 cw, wait 2, land). **Special Missions** are pre-built FSMs that operate in arena coordinates with collision avoidance between drones.

Scan all ArUco markers

Fleet-wide sequential scan. Each drone claims a marker (shared map prevents collisions), approaches head-on at **hover_distance_m**, hovers for **Hover s**, then chains to the next unclaimed marker. Skew and approach tolerances control when APPROACH transitions to HOVER (§18).

UI control	What it does
Target markers	**1-12** default. Use dashes or comma lists. 0 excluded (origin).
Hover s	Dwell time per marker (default 1.5 s).
Approach tol (m)	Distance-error tolerance for APPROACH→HOVER.
Skew tol	Perpendicularity tolerance (~0.12 ≈ 9°).

Capture enemy targets (SDC26)

Waypoint navigation to each opposing team's box, hovering with the camera aimed at the arena centre so many markers stay in view for localization. Six boxes default (3 red + 3 blue), configurable.

UI control	What it does
Target boxes (JSON)	List of **`{id,x,y,home_team}`**.
Home XY	World-frame return point (post-capture).
Face XY	World-frame point the camera faces during transit.
Altitude (m)	Hover altitude above each box (default 1.5).
Hover s	Must be ≥ 2 s (SDC26 capture-hold rule); default 4.

16 Arena Configuration

Marker layout editor — the ground truth consumed by the Position Tracker and Missions. Collapsed by default; click **Show** to expand.

UI control	What it does
Arena width / depth (m)	Physical dimensions (20 × 10.8 for SDC26).
Height min / max (m)	Ceiling + floor clamps used by the boundary guard.
Marker size (m)	Physical side length (default 0.5).
Marker table	Per-marker row: ID · X · Y · Z · wall (front/back/left/right). Click **+ Add marker**, **Delete** on a row, **Save Config**, or **Reset to Defaults**.

17 Tuning Parameters panel

The single home for **every** live-tunable knob. Two columns:

UI control	What it does
Observer PD (left)	22 sliders driving the visual-servo PD loop. Used by every mission — these are the "mission approach" parameters (flight speed / rotation / aggressiveness).
Position Tracker (right)	16 inputs driving arena-frame pose fusion. Split into three rows: top (Profile / FOV / Latency / IMU blend), Filters (Kalman / marker size / Top-K / Outlier), and Precision (pose-hold / blend / Kalman Q/R).

Every knob has a small ■ info-icon next to its label. Click it for a pop-up with the parameter's full explanation, units, and tuning hints. This is the live equivalent of the tables on the next two pages.

18 Parameter reference — Observer PD

These knobs drive the visual-servo PD loop that every mission uses to hover in front of markers. Applied live via **`POST /proxy/aruco/params`**.

Key / Label	Default	Range	Purpose
Hover distance `hover_distance_m`	2.0	0.5–4.0 [metres]	Target stand-off distance from the marker during hover. The mission flies forward until the marker is this far away, then holds. Smaller = closer / more detail but less safety margin against the net. 2 m is the rules-safe default for SDC26.
Approach speed (forward) `fb_max`	45	0–100 [% RC]	Upper clamp for forward throttle when approaching a marker. Scales the P gain output — higher = faster approach but larger overshoot. Pair with dist_p: the effective command is *min(fb_max, dist_p · err_dist)*.
Retreat speed (backward) `fb_back_max`	45	0–100 [% RC]	Upper clamp for backward throttle when the drone is too close. Set lower than fb_max — retreats tend to happen near the net and should be cautious.
Approach aggressiveness `dist_p`	20	0–60 [P · err_dist(m)]	Proportional gain turning distance error (m) into forward RC%. Higher values react more sharply. Start low (10–20) and raise until hover distance is reached without overshoot.
EMA smoothing (α) `ema_alpha`	0.3	0.05–0.95 [0..1]	First-order low-pass on camera-derived errors. 1.0 = no smoothing (fastest, jitteriest); 0.05 = heavy smoothing (laggy). Typical 0.25–0.5.
Yaw/lateral dead-band (err_x) `deadband_x`	0.05	0–0.30 [normalised]	Below this threshold, yaw + strafe commands are zero. Stops the drone hunting around an already-centred marker. Typical 0.03–0.08.
Altitude dead-band (err_y) `deadband_y`	0.06	0–0.30 [normalised]	Below this threshold vertical command is zero. 0.05–0.1 typical.
Skew dead-band `deadband_skew`	0.04	0–0.30 [normalised]	Below this threshold the perpendicular-alignment (strafe) command is zero — small marker tilt doesn't need correcting.
Distance dead-band `deadband_dist_m`	0.2	0–1.0 [metres]	Below this distance error the forward/back command is zero. Keeps the drone parked once it's within range of hover_distance_m.
Yaw P-gain `yaw_p`	10	0–50 [per err_x]	Proportional gain from horizontal image-error to yaw RC%. Higher = snappier rotation but more overshoot.
Lateral P-gain `skew_p`	25	0–50 [per skew]	Proportional gain from marker-tilt to sideways RC%. Drives the strafe that produces a head-on approach.
Altitude P-gain `alt_p`	25	0–100 [per err_y]	Proportional gain from vertical image-error to vertical RC%.
Yaw D-damping `d_yaw`	0.4	0–2 [per °/s (gyro)]	Derivative damping on yaw using the gyro. Cancels oscillation — if the drone visibly orbits the marker, increase.
Lateral D-damping `d_lr`	0.4	0–2 [per cm/s (vgy)]	Derivative damping on sideways RC using the body-frame Y velocity. Prevents over-strafing.
Vertical D-damping `d_ud`	0.5	0–2 [per cm/s (vgz)]	Derivative damping on vertical RC using the body-frame Z velocity.
Fwd/back D-damping `d_fb`	0.4	0–2 [per cm/s (vgx)]	Derivative damping on forward/back RC using body-frame X velocity. Most important D term for not slamming into the net. Combined with the boundary guard this is the primary brake during fast approaches.
Clamp · max yaw `yaw_max`	15	0–80 [% RC]	Hard upper limit for yaw RC%. 20–30 typical — keeps the drone from spinning wildly on large errors.
Clamp · max lateral `lr_max`	20	0–100 [% RC]	Hard upper limit for sideways RC%. 20–40 typical.

Key / Label	Default	Range	Purpose
Clamp · max vertical `ud_max`	25	0–100 [% RC]	Hard upper limit for vertical RC%. 20–40 typical.
RC dead-floor `rc_min`	2	0–10 [% RC]	Below this magnitude the RC output is forced to zero. Anafi ignores very small RC values anyway — this prevents buzzing while hovered.
Cam H-FOV (drawing only) `cam_hfov_deg`	69	30–110 [degrees]	Used ONLY to draw the camera cone in the top-down view — does NOT affect PnP or control. 69° is the Anafi nominal.
Observer marker size `marker_size_m`	0.5	0.05–2.0 [metres]	Physical marker side length for the observer's own PnP. Must match the printed markers. SDC26 markers are 0.5 m. Keep in sync with the Position Tracker's marker size.

19 Parameter reference — Position Tracker

These knobs drive the multi-marker ArUco + IMU fusion that produces the arena-frame pose. Applied live via **`POST /proxy/position/config`**. Values mirror the module constants in **`controller_unified/ctrl_position.py`**.

Key / Label	Default	Range	Purpose
Detection profile `detect_profile`	Balanced	Balanced / Sensitive / Strict	Preset for the ArUco detector parameters. Sensitive catches distant / partially-lit markers at higher CPU cost; Strict rejects noisy detections; Balanced is the default.
Camera H-FOV `fov_deg`	69	40–120 [degrees]	Horizontal field-of-view used to synthesise the intrinsics matrix when no calibration file is loaded. Anafi 4K ≈ 69°. Uploading a .npz calibration overrides this.
Video-to-IMU latency `latency_ms`	200	0–800 [ms]	How old the camera frame is relative to the current IMU sample. The positioner rewinds the IMU buffer by this amount. Use the header Latency row with auto-set enabled to have the C2→FC + FC→drone + video decode total pushed here automatically.
IMU ↔ ArUco blend `imu_weight`	0.30	0–1	Mix between pure ArUco pose (0) and pure IMU dead-reckoning (1). Higher = smoother but drifts more during marker outages. 30% is a good default.
Kalman filter `enable_kalman_filter`	ON	on / off	Per-axis 1-D Kalman on x/y/z. Smoother and handles brief dropouts; OFF means pose jumps straight to the last ArUco solution — noisier but zero added latency.
Marker size `marker_size_m`	0.5	0.05–2.0 [metres]	Physical marker side length for PnP. Overrides arena_config.json when set. SDC26 markers are 0.5 m.
Top-K markers `top_k_markers`	0 (auto=4)	0–10	Use only the N closest markers in the weighted-mean fusion. 0 = auto (4). Smaller K = faster, less robust. Larger K = more samples but includes less-accurate distant detections.
Outlier reject distance `outlier_reject_m`	2.5	0.1–20 [metres]	Per-marker poses further than this from the weighted-mean position are rejected before re-averaging. Tighten for a tidy small arena; loosen if markers are far apart.
Pose hold (dead-reckon) `pose_hold_sec`	0.8	0–10 [seconds]	After the last valid ArUco fix, keep publishing pose via IMU dead-reckoning for this many seconds. Too short → pose vanishes every blink; too long → stale pose drifts metres. 0.5–1.0 s typical.
Minimum reference markers `min_ref_count`	1	1–12	Require at least this many markers visible before a fused pose is accepted. 2–3 gives a much more robust fix by cross-checking.
Minimum reference weight `min_ref_weight`	0.0	0–1	Require the best-matching marker to have at least this fused weight. Rejects very low-confidence fits. 0.2–0.4 is restrictive but clean.
Measurement blend — low `meas_blend_min`	0.35	0–1 [α]	Minimum EMA α applied to fresh ArUco measurements. Used when fix quality is high (trust the filter). Lower = more Kalman smoothing.
Measurement blend — high `meas_blend_max`	0.85	0–1 [α]	Maximum EMA α used when fix quality is low (trust fresh measurements more). The positioner interpolates between min and max based on residual error and ref count.
Velocity blend `vel_blend`	0.25	0–1	Blend between IMU velocity (0) and Kalman-state velocity derivative (1). 0.25 = 25% Kalman-derived, 75% raw IMU. Higher = smoother vz/vy plots but slower reaction.
Max state Δt `max_state_dt`	1.0	0.05–10 [seconds]	If more than this passes between updates, the Kalman state is reset instead of extrapolated. Prevents exploding covariance during outages.
Kalman Q (process var) `kalman_process_var`	1e-3	1e-6–10	How much the state is expected to change between steps. Low Q = smooth/sluggish; high Q = reacts faster but noisier. Try 5e-4 for smooth hover, 5e-3 for dynamic missions.

Key / Label	Default	Range	Purpose
Kalman R (measurement var) `kalman_meas_var`	1e-1	1e-6–10	How noisy the ArUco measurements are. Low R = trust camera (snaps to detections); high R = trust model (smoother but lagging). Large markers at short range can use 1e-2; distant markers 3e-1.

20 Theme switcher

The ■/■■ button in the header toggles between dark (default) and light. Choice is saved in localStorage under **`sdc_theme`**. CSS overrides under **`html[data-theme="light"]`** flip every major colour including panels, buttons, inputs, info-icons, and the parameter-info modal.

21 Info icons (■)

A 13 px circled *i* appears next to every tuning label. Click to open a modal with the parameter's full description. Same text as in this PDF's reference tables — kept in sync by sourcing from the JavaScript **`PARAM_INFO`** map. Click outside the box or the **Close** button to dismiss.

22 Troubleshooting quick-ref

UI control	What it does
Drone bar empty	**`flightctrlN`** hostname not resolving, or the Pi's API isn't running on port 8080.
Takeoff refused	Check the pre-takeoff diagnostic block printed to the Pi server's stdout (alert / motor / sensors / magneto).
Pose jitters > 10 cm	Verify **marker_size_m = 0.5** in both the observer and the position tracker; upload a real calibration .npz; raise **imu_weight**; turn on Kalman.
Drone stutters during approach	Increase **fb_max** slowly, not in one jump. Keep RC tick at 400 ms default.
Drone flies too fast into net	Lower **fb_max** and/or **dist_p**; the boundary guard uses velocity to predict the next 0.35 s — ensure telemetry is live ("Vel" in coord readout is not dashes).
Mission won't rotate after hover	Confirm **search_rc** is non-zero during SEARCH by inspecting the JSONL in **`controller/logs/`**.
Latency widget stays red	**fc** → **drone** ping timing out — switch Wi-Fi channel or band in the Anafi panel.